



AI-Based Battery Health Analytics Platform

Team ID: AX-302

Track: AI

Team name: Shadow Syndicates

Team members: B.Vishnuvardhan,
F.Kevin Cris, P.John Ezra,
B.Kamaleshwaran.

PROBLEM STATEMENT

Core problem: Battery degradation is hard to estimate accurately, leading to unexpected failures, costly downtime, and poor lifecycle management. Manual monitoring is imprecise and offers no predictive insight.

Significance: Batteries power everything from EVs to grid storage — undetected degradation causes safety risks, unplanned replacements, and wasted energy. Aligns with SDG 7 (Affordable & Clean Energy) and SDG 9 (Industry, Innovation & Infrastructure).

Target users: EV owners/fleets, renewable energy operators, battery manufacturers, industrial equipment maintainers.

Gaps in current solutions: Existing monitoring is reactive (fails after damage occurs), lacks predictive modeling, provides no actionable charging recommendations, and has no centralized health logging.

METHODOLOGY PROPOSED

Overview: An AI-based predictive analytics platform that ingests battery usage data (voltage, current, temperature, charge/discharge cycles) and outputs State-of-Health (SoH), Remaining Useful Life (RUL), and charging efficiency insights.

How it solves the problem: Replaces manual monitoring with a trained ML model that detects degradation trends early, flags anomalies (overcharging, rapid wear), and proactively schedules replacement/maintenance.

Innovative features: Timestamped health-report logging, RUL prediction with end-of-life alerts, anomaly detection for charging behavior, actionable recommendations (not just raw metrics) — presented through a clean, real-time dashboard.

Assigned bonus feature: AI smart search Using LLM

TECHNICAL APPROACH

Architecture: Data (battery usage dataset) → Preprocessing/cleaning → Train a model using XGboost library(python) via LoRA → Inference layer → Streamlit dashboard for visualization.

Tools/technologies: Python, PEFT (LoRA), Pandas, Scikit-learn (for SoH/RUL regression if tabular), Streamlit (UI), Google Colab (training/GPU).

Workflow: Load and clean battery dataset → Train model on historical usage data → Validate predictions against test data → Deploy inference in Streamlit → Log outputs with timestamps.

Data management: Dataset sourced from Kaggle/NASA Battery dataset , split into train/test, cleaned for missing/noisy sensor values.

Testing/metrics: Model evaluated using RMSE/MAE for SoH and RUL prediction accuracy; anomaly detection validated against known overcharge/degradation patterns in test data.

FINAL REFLECTIONS AND FUTURE STEPS

Software tools/frameworks: Python, PEFT, Scikit-learn, Streamlit, Pandas, NumPy.

Hardware: Standard laptop; GPU via Google Colab (no additional hardware required for prototype).

Deployment platform: Google Colab (training), Streamlit Community Cloud or local hosting (demo).

Programming languages: Python.

Future steps: Expand to cloud-based fleet-wide monitoring, integrate with renewable energy systems, real-time IoT sensor data ingestion for live monitoring at scale.

EXPECTED DELIVERABLE

A working Streamlit web application where users upload battery usage data and receive: SoH score, predicted Remaining Useful Life, charging efficiency analysis, and anomaly alerts — all logged with timestamps.

EXPECTED DELIVERABLE

Research paper / GitHub repository with README documenting problem approach, model architecture, dataset source, training methodology (LoRA & Unsloth), and evaluation metrics.

Documentation

EXPECTED DELIVERABLE

Live Streamlit demo showing real-time input → SoH/RUL prediction → charging efficiency recommendation, plus a backup recorded demo video.

REFERENCES

Research Publications

Ahwiadi, M. & Wang, W. (2025). Battery Health Monitoring and Remaining Useful Life Prediction Techniques: A Review of Technologies. Batteries, 11(1), 31. — mdpi.com/2313-0105/11/1/31

Official Documentation

XGBoost — xgboost.readthedocs.io

Scikit-learn — scikit-learn.org

SHAP — shap.readthedocs.io

FastAPI — fastapi.tiangolo.com

Streamlit — docs.streamlit.io

Datasets:

NASA PCoE Battery Dataset — nasa.gov/intelligent-systems-division/discovery-and-systems-health/pcoe/pcoe-data-set-repository

CALCE Battery Data, University of Maryland — calce.umd.edu/battery-data

-All datasets, tools and references used are open-source/publicly available and cited for academic purposes only

THANK YOU